

LOW-COST DRY-CONTACT SURFACE EMG ACQUISITION DEVICE FOR WIRELESS CONTROL APPLICATIONS

Nadarasar Bahavan
Gevindu Wickramaarachchi
Kithmin Wickremasinghe

University of Moratuwa, Sri Lanka

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Surface Electromyography

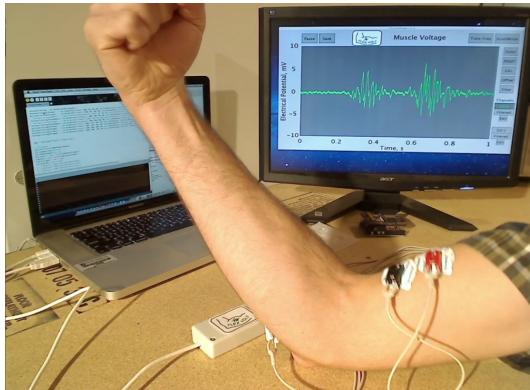


Figure: Surface electromyography (sEMG) is technique through which a bioelectrical signal generated from muscular contractions and relaxations is acquired from the surface of the skin.

Applications

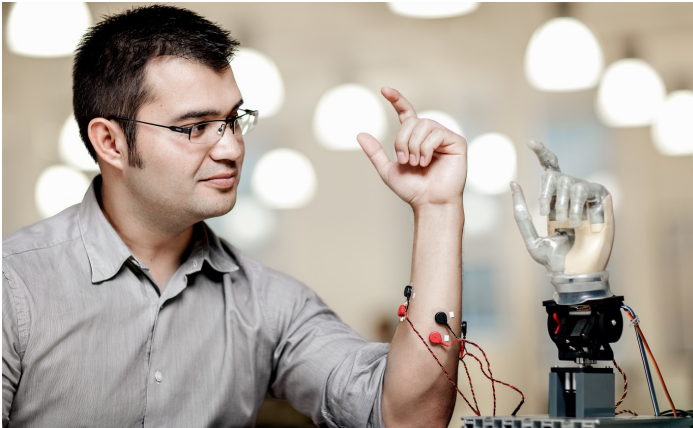


Figure: It has various applications in haptics, wearables, bionics and gesture detection.

Sensors

- Dry-contact electrode devices are currently more preferred over wet electrodes due to the durability, ease-of-use and re-usability
- In this study, an sEMG acquisition device was designed as a neural interface using dry stainless steel electrodes and an Adafruit Feather M0 WiFi board (Adafruit Industries, USA). Bipolar electrode configuration was chosen for its better focal sEMG acquisition area.
- The sensor used for the device has been designed according to the SENIAM (Surface EMG for Non-invasive Assessment of Muscles) standards and specifications described in Naim et. al [1].

Architecture

- The signal acquired from the bi-polar electrodes were transmitted to the acquisition circuit via single core shielded wires to prevent power-line noise contamination via capacitive coupling.
- Then, the signal was amplified using an using AD623AN (Analog Devices, USA) instrumentation amplifier to the dynamic range of the analogto- digital converter (ADC), before being band-passed filtered within the range of 60 Hz to 150 Hz to remove artifacts and noise.
- The band-pass filter was implemented with an LMP7702 using the butterworth configuration. The conditioned signal was acquired by the Feather Board at a sampling rate of 256 Hz which transmitted the signal via a WiFi signal to a computer for further analysis.

Measures to prevent Motion Artifacts



Figure: There is a tendency to generate motion artifacts while using the device which was designed to be wireless and portable. These artifacts materialize due the impedance variations between the skin-electrode interface due to unstable contact. To avoid that, the bipolar electrode is held tightly in place using a strap.

BIOPAC based study

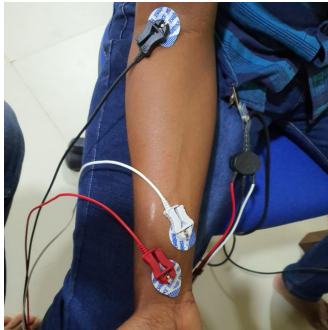


Figure: Additionally, since there is only a single bipolar electrode pair, it is essential to select a location on the arm which can characterize the selected finger movements adequately. Hence, after performing initial experiments with a BIOPAC System (BIOPAC Systems, Inc), the bottom side of the forearm was chosen.

Circuit Simulation

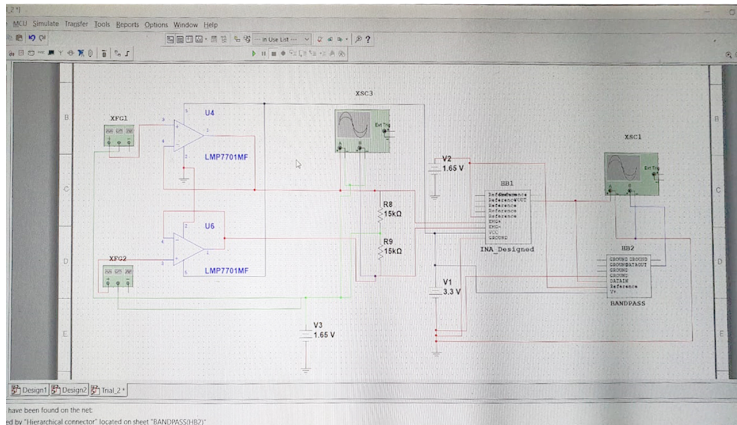


Figure: The Computer Simulation was first implemented using the software Multisim (National Instruments)

Common Mode Rejection Ratio

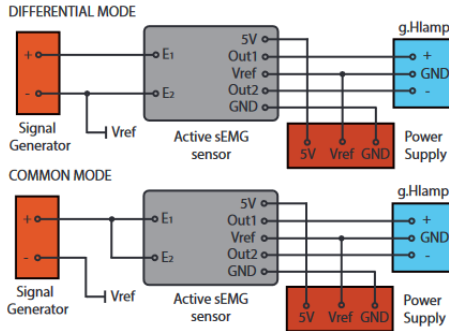


Figure: In order to characterize the Common Mode Rejection Ratio CMRR of the circuitry, the bi-polar electrodes were connected to a common end (+) of a signal generator and while the other end of the signal generator (-) was grounded as specified in Naim et. al [1].

Testing: Oscilloscope

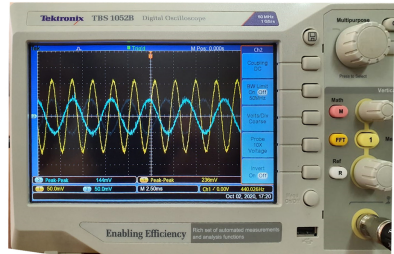


Figure: FFT based characterization of the outputs: The resulting conditioned signal of the circuitry was observed through an Oscilloscope. A signal generator was used to generate various sine waves of different frequencies. Fast fourier transform (FFT) allowed to fully characterize the common mode response. The differential mode response can be obtained by connecting different signals to the electrodes.

Device

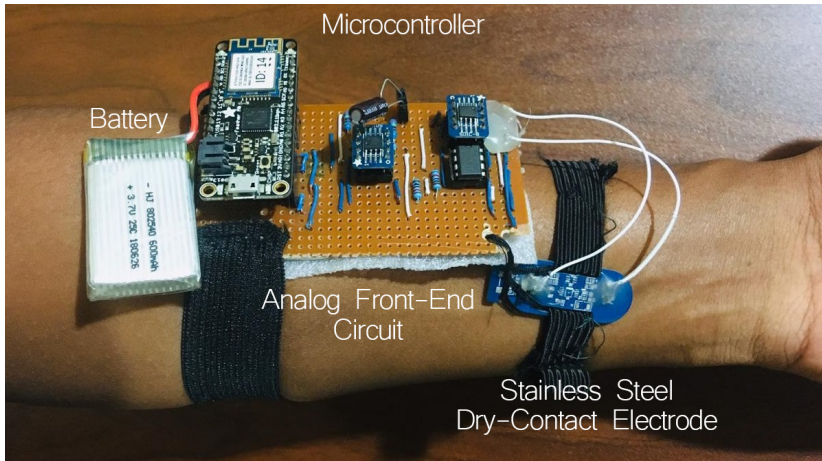


Figure: Internal circuitry

Computer Simulation

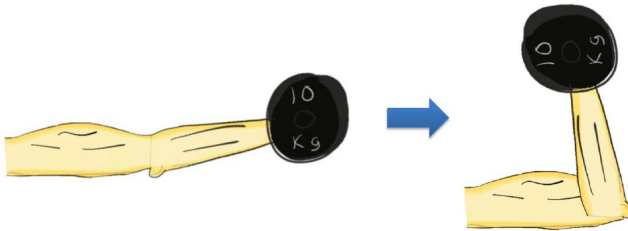


Figure: Computer Simulation

The computer simulation was implemented using python libraries to showcase that the device can be used to demonstrate a wireless control application where work is done, such as lifting a weight.

Conclusions

The low cost neural machine interface developed in this study is used as a control device to perform mechanical work using an artificial prosthetic hand, which is demonstrated using a computer simulation. Prosthetic arm controlling using sEMG based hand gesture recognition is one of the main control application where sEMG can be used [2]. Such neural machine interfaces can be developed using sEMG to be used for diverse control applications, from VR gaming to freehand typing and even sign language translation

References

- ① A. M. Naim, K. Wickramasinghe, A. De Silva, M. V. Perera, T. D. Lalitharatne, and S. L. Kappel. Low-cost active dry-contact surface EMG sensor for bionic arms. In 2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC), pages 3327–3332, 2020.
- ② S. Benatti, B. Milosevic, F. Casamassima, P. Sch onle, P. Bunjaku, S. Fateh, Q. Huang, and L. Benini. EMG-based hand gesture recognition with flexible analog front end. In 2014 IEEE Biomedical Circuits and Systems Conference (BioCAS) Proceedings, pages 57–60, 2014.